

- 5 (a) Insulin has an important role in the maintenance of blood glucose concentration.

An investigation measured how blood glucose concentration and blood insulin concentration changed after a person had eaten a glucose-rich meal.

The results are shown in Figure 5.1.

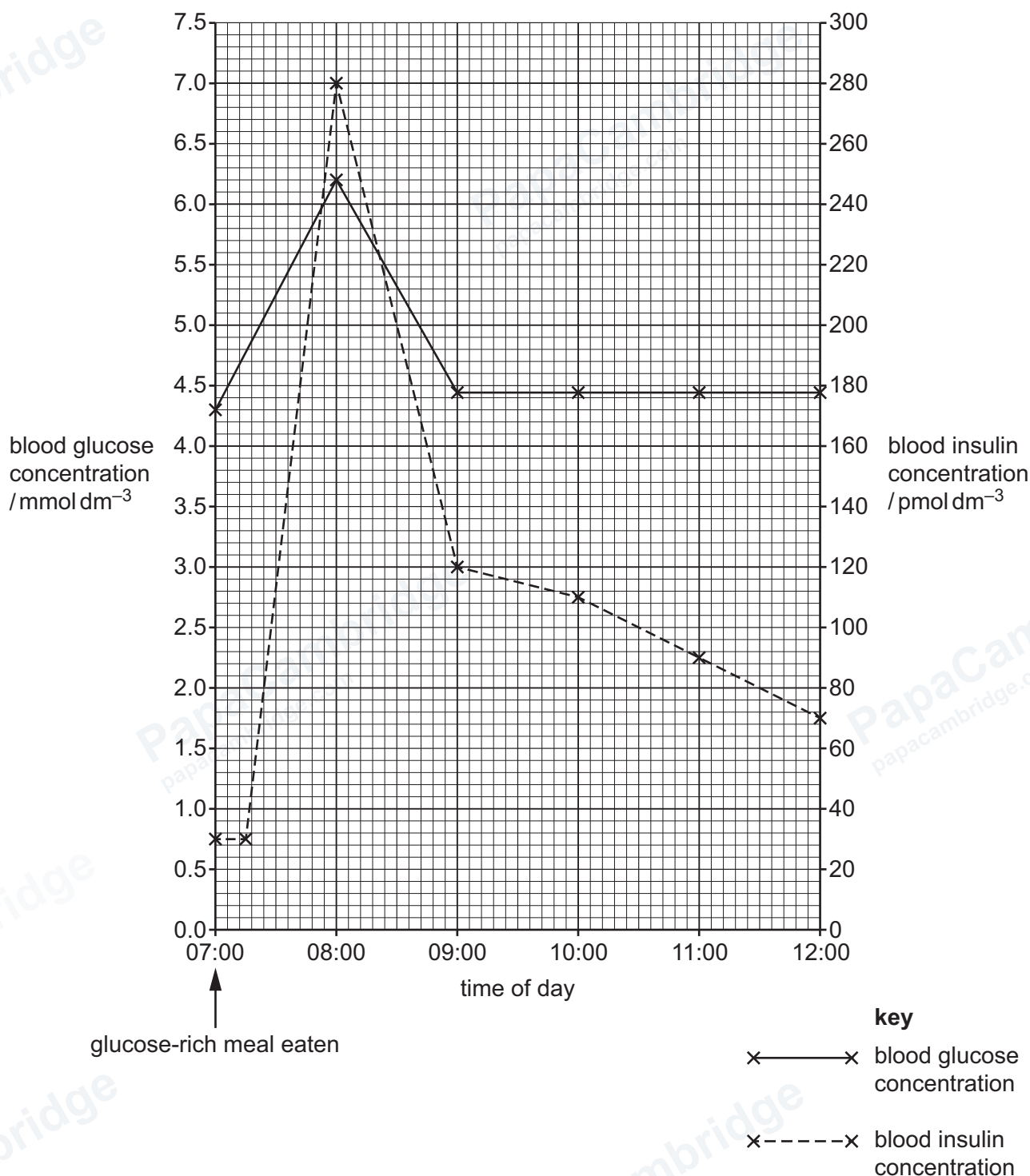


Figure 5.1



- (i) Use Figure 5.1 to calculate the percentage increase in the blood **insulin** concentration between 07:00 and 08:00.

Give your answer to **four** significant figures.

..... % [2]

- (ii) With reference to Figure 5.1, explain the change that occurs in the blood glucose concentration between 08:00 and 09:00.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (b) Type 1 diabetes in humans is a condition in which the pancreas does not produce enough insulin to control blood glucose concentration. People with type 1 diabetes measure their blood glucose concentration at regular intervals using either a biosensor or test strips.

- (i) Name the **two** enzymes that can be used in biosensors and test strips to measure blood glucose concentration.

.....

..... [2]



- (ii) People with type 1 diabetes need daily injections of insulin. Insulin used to be obtained by extracting it from farm animals. Now it is made artificially using genetic technology and has the same amino acid sequence as human insulin.

Explain the advantages of producing insulin, with the same amino acid sequence as human insulin, by genetic technology rather than obtaining it from farm animals.

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 11]